COMP364 Quiz 5

Question 1. Describing the value-learning algorithm (5 points)

(a) State the update equation of the value-learning algorithm studied in class.

Solution.

$$V(S_t) \leftarrow V(S_t) + \alpha\big(V(S_{t+1}) - V(S_t)\big)$$

(b) What phrase would a reinforcement learning expert use to describe our value-learning algorithm?

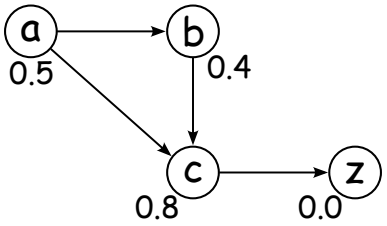
Solution.

It is a "tabular solution using temporal difference updates."

Question 2. Implementing the value-learning algorithm (10 points)

Show the outputs of two episodes of our value-learning algorithm for the two-player game shown below, assuming:

- the start state is a, and the final state is z;
- the current estimated values are as shown on the graph;
- $\epsilon = 0.3$ and $\alpha = 0.1$;
- the random numbers used by the algorithm, if needed, are $\rho_1 = 0.88, \rho_2 = 0.21, \rho_3 = 0.62, \rho_4 = 0.89, \rho_5 = 0.48$.



You must use the notation and method discussed in class.

Solution.

episode 1	a	$\rho_1 = 0.88 \xrightarrow{B}$	b	$\xrightarrow{B}$	c	$\xrightarrow{B}$	z
	0.50		0.40		0.80		0.00
	0.51		0.38		0.82		

episode 2	a	$\rho_1 = 0.21 \xrightarrow{E}$	c	$\xrightarrow{B}$	z
	0.51		0.82		0.00
			0.838		

Total number of points: **15**